

CHANGTING, China

WMO: 589110

Lat: 25.8527N

Lon: 116.3659E

Elev: 311

StdP: 97.64

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	0.9	-6.3	2.3	6.0	-3.9	2.8	5.3	6.1	7.4	5.4	8.1	0.8	320	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	34.6	25.1	33.7	25.0	32.7	24.8	26.3	31.5	25.9	31.0	25.6	30.4	2.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	21.1	28.2	24.7	20.5	27.8	24.3	20.0	27.5	84.3	31.4	82.4	31.0	80.8	30.7	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	5.4	4.5	3.9	DB	-3.2	36.3	1.6	0.8	-4.3	36.8	-5.2	37.3	-6.1	37.7	-7.3	38.3
(5)				WB	-3.9	27.2	1.6	0.6	-5.1	27.7	-6.1	28.1	-7.0	28.4	-8.2	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	9.2	11.5	14.9	19.9	23.5	26.2	28.0	27.7	25.4	21.2	16.2	10.8
	DBStd	7.35	4.27	5.13	4.24	3.46	2.48	1.92	1.37	1.51	2.52	2.93	3.76	3.78
	HDD10.0	157	69	41	8	1	0	0	0	0	0	0	3	35
	HDD18.3	955	284	197	122	24	1	0	0	0	0	8	82	236
	CDD10.0	3654	43	83	159	298	418	485	558	548	463	348	190	59
	CDD18.3	1411	1	6	15	72	161	235	300	290	213	98	19	2
	CDH23.3	12141	2	35	83	390	1057	1973	3164	2864	1787	658	124	4
CDH26.7	4503	0	3	7	77	310	681	1389	1177	679	164	16	0	
Wind	WSAvg	1.4	1.5	1.5	1.4	1.4	1.2	1.3	1.4	1.4	1.4	1.5	1.5	1.5
Precipitation	PrecAvg	1734	59	98	190	231	285	304	120	149	107	61	46	42
	PrecMax	2452	176	281	400	562	524	671	277	370	253	193	160	168
	PrecMin	1244	2	0	31	65	73	16	12	27	7	0	0	1
	PrecStd	286	55	61	100	108	91	152	62	84	64	52	38	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	26.7	27.6	30.6	32.7	34.2	36.1	35.6	34.4	31.9	28.5	23.8
		MCWB	16.4	18.9	19.5	21.5	22.6	25.3	24.7	25.1	25.0	23.2	19.9	18.1
	2%	DB	20.8	24.2	25.7	28.8	31.3	32.9	34.7	34.6	33.2	29.9	26.5	21.3
		MCWB	15.2	17.6	18.8	21.3	22.6	24.7	25.1	25.2	24.7	21.6	19.0	15.5
	5%	DB	18.6	22.1	23.8	27.3	29.9	31.8	33.8	33.5	32.0	28.5	24.6	19.2
		MCWB	13.7	16.7	17.7	20.6	22.6	24.7	25.1	25.2	24.4	20.7	17.7	14.0
	10%	DB	16.2	19.4	21.7	25.7	28.4	30.5	32.7	32.2	30.6	27.1	22.7	17.2
		MCWB	12.7	15.3	17.2	20.1	22.5	24.5	25.0	25.1	23.7	20.1	16.9	13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	20.3	21.3	23.5	25.7	26.7	26.9	26.8	26.3	24.4	22.0	19.8
		MCDB	21.3	24.8	25.5	27.4	29.5	31.2	31.7	32.7	31.9	29.6	26.4	22.3
	2%	WB	16.3	18.7	20.2	22.6	24.9	26.0	26.2	26.2	25.6	23.3	20.7	17.1
		MCDB	19.5	22.8	23.7	26.5	28.6	30.2	31.3	31.7	30.9	27.5	24.2	19.5
	5%	WB	14.8	17.3	19.2	21.9	24.2	25.5	25.9	25.8	25.0	22.5	19.6	15.5
		MCDB	17.9	21.0	22.2	25.6	27.5	29.6	31.0	31.0	30.2	26.3	22.5	17.8
	10%	WB	13.1	15.9	18.1	21.3	23.6	25.1	25.5	25.4	24.5	21.7	18.4	14.0
		MCDB	15.7	19.0	20.7	24.4	26.5	29.0	30.4	30.3	29.0	25.3	21.2	16.4
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.2	8.0	8.1	7.8	7.2	8.9	8.3	8.7	9.8	10.1	9.8
		MCDBR	13.2	12.7	12.3	11.4	11.4	9.9	11.0	10.8	10.8	12.4	13.5	13.8
		MCWBR	7.3	6.4	5.6	4.6	4.1	3.0	2.9	3.0	3.2	4.7	6.1	7.5
	5% WB	MCDBR	10.5	10.8	9.1	8.2	7.5	7.8	9.2	9.2	9.2	9.0	9.2	9.2
		MCWBR	6.2	5.8	5.0	3.9	3.2	2.9	3.0	3.0	3.2	3.8	4.7	5.7
Clear-Sky Solar Irradiance	taub	0.604	0.659	0.761	0.739	0.622	0.583	0.556	0.612	0.618	0.628	0.570	0.554	
	taud	1.668	1.606	1.464	1.548	1.815	1.923	1.998	1.834	1.792	1.708	1.789	1.781	
	Ebn at Noon	632	637	602	636	715	738	758	715	695	652	658	653	
	Edh at Noon	219	249	302	284	217	193	179	210	214	222	192	188	
All-Sky Solar Radiation	RadAvg	2.39	2.63	2.85	3.34	3.77	4.02	5.34	4.82	4.26	3.78	2.94	2.69	
	RadStd	0.50	0.56	0.47	0.53	0.58	0.46	0.48	0.41	0.40	0.63	0.45	0.54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.42	N/A	N/A	+0.75	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.32	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	+119	+87

Nomenclature: See separate page